SGA 6 - Expose XIV

Sections 3.3 (continuation) through 4.7

Source coordinates: current-rescribe idx685-idx692; printed pages 672-679; source-PDF pages 675-682. The opening repeats only the minimal idx684 carry-in required to complete the sentence across the page break.

3.3 (continuation). This holds at least when Y and X are both smooth over a scheme S, by the theorem of relative cohomological purity (SGA 4 XVI 3.7); conjecturally, it also holds when Y and X are both regular, by the theorem of absolute cohomological purity, proved at least for excellent X of characteristic zero (SGA 4 XIX 3.2).

When X and Y are smooth over the field of complex numbers, one may also use the integral Chern class on the locally compact space $X(\mathbb{C})$ of complex points of X. Formula (3.1) was proved in this case by Atiyah and Hirzebruch [1], who more generally treat embeddings of complex analytic varieties (working with naive K'). It follows, by the Lefschetz principle and the general results of SGA 4, that the ℓ-adic formula (3.1) holds whenever Y and X are smooth over an algebraically closed field of characteristic zero. This makes the cohomological formula (3.1) extremely plausible. Nevertheless, it should be noted that, at present, it has not been proved even when X and Y are smooth over a field k of characteristic zero, if k is not algebraically closed.

3.4. One may also work with the cohomological theory of Chern classes, in the de Rham or Hodge style [13]. The question of the validity of (3.1) in this setting arises only when X is not of characteristic zero, since one knows in any case (no. 6) that the formula in question is true modulo torsion. For smooth quasi-projective schemes over a field k, the de Rham or Hodge version of (3.1) would moreover follow from the version using cycle classes modulo algebraic equivalence. Thus the de Rham or Hodge cohomological version provides another accessible way to test the validity of the cycle-class version of (3.1).

3.5. The proof [1] of the transcendental cohomological version of (3.1) is essentially transcendental in nature and uses tubular neighborhoods of Y in X. This suggests that a proof of a formula of type (3.1) may be tied to the systematic consideration of $K^\bullet$-groups and cohomology groups with support in Y, which might provide an adequate technical means of *localization around* Y.

4. Relations between $K^\bullet(X)$ and the Chow ring $A(X)$

4.1. Suppose for simplicity that X is a smooth quasi-projective scheme over a field k (see no. 8 for a more general case), so that the Chow ring $A(X)$ of cycle classes modulo rational equivalence is available [4]. Two variants of this ring, defined under appreciably more general conditions, are $\mathrm{Gr}^\bullet(X)$, the graded ring associated with the λ-filtration of $K^\bullet(X)$, and $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, the graded ring associated with the filtration of $K^\bullet(X) \simeq K_\bullet(X)$ by codimension of supports. There are theories of Chern classes with values in each of these three rings (see [9] for the case of $A(X)$, and this seminar for $\mathrm{Gr}^\bullet(X)$, which maps into $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$). There is also an ℓ-adic cohomological theory (SGA 5 VII). Thus, when ℓ is prime to the characteristic of k,

$$c_i(E) \in H^{2i}(X, \mathbb{Z}_\ell(i)),$$

where E is a vector bundle on X, or more generally any element of $K^\bullet(X)$. These theories are related by the following canonical homomorphisms, which are compatible with the multiplicative structures:

$$
\begin{tikzcd}
H^{2*}(X,\mathbb{Z}_\ell(*)) & \\
A(X) \arrow[u, "\gamma"] \arrow[r, "\varphi"] & \mathrm{Gr}^\bullet_{\mathrm{top}}(X) \\
& \mathrm{Gr}^\bullet(X) \arrow[u, "j"']
\end{tikzcd}
\tag{4.1}
$$

where

$$H^{2*}(X,\mathbb{Z}_\ell(*)) = \bigoplus_i H^{2i}(X,\mathbb{Z}_\ell(i)).$$

The homomorphism j comes from the fact that the λ-filtration of $K^\bullet(X)$ is finer than the topological filtration by codimension of support (Expose X), and it is an isomorphism modulo torsion (Expose VII 4.11). The homomorphism φ, defined for example in [RRR], associates to the class of the codimension-i cycle carried by a closed subscheme $Y \subset X$ the element $\mathrm{cl}^i(\mathcal{O}_Y)$ of

$$\mathrm{Filt}^i_{\mathrm{top}} K^\bullet(X)/\,\mathrm{Filt}^{i+1}_{\mathrm{top}} K^\bullet(X).$$

It is surjective, and we shall see in 4.2 that it is an isomorphism modulo torsion. Hence, modulo torsion, the three rings $A(X)$, $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, and $\mathrm{Gr}^\bullet(X)$ are canonically isomorphic; consequently, the Riemann–Roch formulas expressed using any one of these rings after tensoring with $\mathbb{Q}$ are equivalent.

The graded ring $\mathrm{Gr}^\bullet(X)$, used in this seminar chiefly as a substitute for the Chow ring, has the considerable advantage of being defined for every locally ringed topos, and in particular for every scheme, whether regular or not. Its disadvantage (4.6) with respect to $A(X)$ and $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ is that it is not covariant as it stands for proper morphisms, even between smooth quasi-projective schemes over a field; it acquires that covariance only after tensoring with $\mathbb{Q}$.

Another advantage of $A(X)$ over its competitors $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ and $\mathrm{Gr}^\bullet(X)$ is the homomorphism

$$A(X) \longrightarrow H^{2*}(X,\mathbb{Z}_\ell(*)),$$

which permits cohomological Chern classes to be interpreted as images of Chern classes with values in $A(X)$. In general this homomorphism does not factor through $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ (4.7), and it is at least doubtful that there is in general a homomorphism

$$\mathrm{Gr}^\bullet(X) \longrightarrow H^{2*}(X,\mathbb{Z}_\ell(*))$$

carrying Chern classes to Chern classes (see no. 5). These difficulties disappear after tensoring with $\mathbb{Q}$, and one obtains a homomorphism

$$\mathrm{Gr}^\bullet_{\mathrm{top}}(X) \longrightarrow H^{2*}(X,\mathbb{Q}_\ell(*)) = H^{2*}(X,\mathbb{Z}_\ell(*)) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell \tag{4.2}$$

that carries Chern classes to Chern classes.

4.2. To show that the surjective homomorphism φ in (4.1) is an isomorphism modulo torsion, one uses the Chern homomorphism obtained from the theory of Chern classes [9],

$$K^\bullet(X) \longrightarrow \mathrm{Ch}(A(X)),$$

where the target is the Chern ring of $A(X)$ as defined in Expose V. This homomorphism is compatible with the topological filtration of $K^\bullet(X)$ and the natural filtration of $\mathrm{Ch}(A(X))$, and therefore induces a homomorphism of graded groups

$$\psi : \mathrm{Gr}^\bullet_{\mathrm{top}}(X) \longrightarrow A(X). \tag{4.3}$$

One proves the formulas

$$\begin{cases} \psi\varphi(x) = (-1)^{i-1}(i-1)!\, x \qquad (\mathrm{mod\ torsion}), \quad x \in A^i(X), \\ \varphi\psi(x) = (-1)^{i-1}(i-1)!\, x \qquad (\mathrm{mod\ torsion}), \quad x \in \mathrm{Gr}^i_{\mathrm{top}}(X). \end{cases} \tag{4.4}$$

The first formula implies the second because φ is surjective. These formulas show that φ and ψ are isomorphisms modulo torsion and, up to the factor $(-1)^{i-1}(i-1)!$ in degree i, are inverse to one another.

To prove the first formula in (4.4), one may suppose that x is represented by an irreducible closed subscheme Y of X of codimension i. The formula is then equivalent to

$$c_i(\mathcal{O}_Y) = (-1)^{i-1}(i-1)!\ \mathrm{cl}(Y) \qquad (\mathrm{mod\ torsion}). \tag{4.5}$$

On the left, $\mathcal{O}_Y$ is regarded as a coherent sheaf on X and hence defines an element of $K^\bullet(X)$; on the right, $\mathrm{cl}(Y)$ denotes the class of Y in $A^i(X)$. Since $A^i(X)$ is unchanged upon deleting from X a closed subset of codimension $> i$, one reduces to the case where Y is regular, and therefore smooth over the base field k if k is perfect. In that case, (4.5) is precisely the Riemann–Roch formula for the inclusion $Y \to X$ and the element $1 \in K^\bullet(Y)$, written using Chern classes with values in the Chow ring $A(X)$. Under the present hypotheses, this formula is proved in [2] by a method very close to the one used in this seminar for the analogous formula with values in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$.

Thus, at least when k is perfect, (4.4) follows, and so does the fact that φ is an isomorphism modulo torsion. This in turn implies, a posteriori, that Riemann–Roch with denominators for a proper morphism of smooth algebraic schemes over k is essentially the same whether it is written using $A(X)$, as in [2], or using $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, as in this seminar. The case of an arbitrary k reduces easily to the algebraically closed case: for a finite extension k'/k of degree n, the kernel of the natural map

$$A(X) \xrightarrow{f_X^*} A(X \otimes_k k')$$

is annihilated by n. Indeed, the trace homomorphism

$$f_{X*} : A(X \otimes_k k') \longrightarrow A(X)$$

satisfies $f_{X*} f_X^* = n\,\mathrm{id}$.[1]

4.3. It seems quite plausible that formulas (4.4) hold exactly as stated, and not merely modulo torsion; equivalently, that formula (4.5) holds not merely modulo torsion. This question is a very special case of the question whether the denominator-free Riemann–Roch formula (3.1) is valid in $A(X)$ for an immersion $Y \to X$, namely the case $i = d$. This special case may reasonably be expected to hold. Indeed, we already noted in no. 3 that, when k has characteristic zero, the denominator-free Riemann–Roch formula is valid at least in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$. Consequently the two sides of (4.5) have the same image in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$; equivalently, the second formula in (4.4) holds. This result will extend automatically to arbitrary characteristic once

[1]Thus the formula is established, thanks to P. Jouanolou, for X smooth and quasi-projective over a field (see the note on p. 5).

the question raised in no. 1 has been satisfactorily resolved. As evidence in support of formula (4.5) in $A(X)$ itself, let us note that it is clear a priori that the restriction of the left-hand side to $X - Y$ is zero. Thus, since Y is assumed irreducible, the left-hand side must a priori be an integral multiple of $\mathrm{cl}(Y)$:

$$c_i(Y) = n(Y)\,\mathrm{cl}(Y).$$

This already proves the exact formula (4.5) whenever $\mathrm{cl}(Y)$ is not a torsion element of $A^i(X)$. It is immediate that this is the case if X is projective (and not merely quasi-projective) over the field k: indeed, if $\xi \in A^1(X)$ is a polarization of X, then $\deg(Y \cdot \xi^i) > 0$. This proves the desired formula in that case. It follows that the formula is still valid whenever X is isomorphic to an open subscheme of a smooth projective scheme over k, as will be the case when an adequate solution of the problem of resolution of singularities is available. In particular, this holds when k has characteristic zero, by Hironaka [11].

The validity of the exact formula (4.5), and hence of (4.4), would imply that the kernel of

$$\varphi^i : A^i(X) \longrightarrow \mathrm{Gr}^i_{\mathrm{top}}(X)$$

is annihilated by $(i-1)!$, and in particular *that* $\varphi^2 : A^2(X) \to \mathrm{Gr}^2_{\mathrm{top}}(X)$ *is an isomorphism.* (That φ^1 is an isomorphism already follows immediately from X 5.3.2.) Unless I am mistaken, the assertion that φ^2 is an isomorphism has not yet been proved, even in the special case $\dim X = 3$. It was therefore in any event premature for the editor to assert the validity of (4.4) and (4.5) in [9, no. 4, 3°].

4.4. At present, the principal advantage of $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ (or of $\mathrm{Gr}^\bullet(X)$) over the Chow ring $A(X)$ is that it is better suited to explicit calculations of certain intersection formulas between cycles in cases of excess intersection, that is, intersections of excessively large dimension. Indeed, the definition of the intersection class by means of Chow's lemma, which allows a cycle to be moved sufficiently within its class, is poorly suited to calculation. A striking example in this regard is the following fundamental formula [Exposé VII 2.7], for a closed immersion $f : Y \to X$ of smooth quasi-projective schemes over k:

$$f^* f_*(y) = y\, c_d(\underline{N}), \tag{4.6}$$

a formula valid in $\mathrm{Gr}^\bullet_{\mathrm{top}}(Y)$ for $y \in \mathrm{Gr}^\bullet_{\mathrm{top}}(Y)$, where $\underline{N}$ is the conormal sheaf

of Y in X, assumed everywhere to have rank d. In the cited passage, this formula follows from a very simple computation of the sheaves $\underline{\mathrm{Tor}}$. By 4.2, the preceding formula implies the same formula in $A(Y)$, but only modulo torsion. It is quite remarkable that formula (4.6) in $A(Y)$ itself has not yet been proved,[2] nor even the special case obtained by taking $y = 1$, which gives the self-intersection of Y:

$$f^*(\mathrm{cl}(Y)) = c_d(\underline{N}), \tag{4.7}$$

even when k is the field of complex numbers.

An analogous problem arises when one attempts to compute the Chow ring of the variety X' obtained by blowing up X along Y, by imitating the method used to compute $\mathrm{Gr}^\bullet(X')_{\mathbb{Q}}$ [Exposé VII 4.8], which also works (without tensoring with $\mathbb{Q}$) for the computation of $\mathrm{Gr}^\bullet_{\mathrm{top}}(X')$. With the notation of the cited passage, the key formula for this computation is

$$f^*(i_*(y)) = j_*(g^*(y) c_{d-1}(\check{\underline{F}})). \tag{4.8}$$

At present, this formula has been established in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X')$ (for $y \in \mathrm{Gr}^\bullet_{\mathrm{top}}(Y)$) by a computation of $\underline{\mathrm{Tor}}$ (Exposé VII 3.4), while its validity in $A(X')$ (for $y \in A(Y)$) remains to be proved.

[2]Cf. the note on p. 23.

Let us note that even the ℓ-adic analogues of formulas (4.6), (4.7), and (4.8) have not yet been proved; when y is the cohomology class of an algebraic cycle, formulas (4.6) and (4.8) are in any event true modulo torsion.

4.5. In this section and the following ones, we shall give examples for which, in diagram (4.1), the homomorphism j, respectively the homomorphism φ, is not an isomorphism. For a given X, it is plainly equivalent to say that

$$j : \mathrm{Gr}^{\bullet}(X) \longrightarrow \mathrm{Gr}^{\bullet}_{\mathrm{top}}(X)$$

is an isomorphism, a monomorphism, or an epimorphism. Moreover, the image of j is the subring of $\mathrm{Gr}^{\bullet}_{\mathrm{top}}(X)$ generated by the Chern classes of locally free sheaves on X. If $K^{\bullet}(X)$ is generated as a $\mathbb{Z}$-algebra by the classes of invertible sheaves on X, one sees at once that the subring in question is also the subring generated by $\mathrm{Gr}^{1}_{\mathrm{top}}(X)$, the degree-one component of $\mathrm{Gr}^{\bullet}_{\mathrm{top}}(X)$.

Now take for X a flag variety

$$X = G/B,$$

where G is a simply connected semisimple algebraic group over the algebraically closed field k, and B is a Borel subgroup [5]. When $k = \mathbb{C}$, the Bruhat cell decomposition of X [5, Exposé 13] implies that the canonical homomorphism

$$K^{\bullet}(X) \longrightarrow K(X(\mathbb{C})) \tag{*}$$

is an isomorphism, as is the canonical homomorphism

$$A(X) \longrightarrow H^{*}(X(\mathbb{C}), \mathbb{Z}), \tag{**}$$

where $X(\mathbb{C})$ is the compact space of complex points of X, endowed with its usual topology induced by that of $\mathbb{C}$. On the other hand, L. Hodgkin [12] proved that $K(X(\mathbb{C}))$ is generated by the classes of invertible sheaves on X. The same is therefore true of $K^{\bullet}(X)$ itself; moreover, it is not difficult to deduce, by a specialization argument using semisimple groups split over $\mathbb{Z}$ (SGA 3), that the result remains valid over every algebraically closed field k. The cell decomposition of X (or, when $k = \mathbb{C}$, the bijectivity of $(**)$) also readily implies that $A(X)$ is torsion-free. Hence, by 4.2, the homomorphism

$$\varphi : A(X) \longrightarrow \mathrm{Gr}^{\bullet}_{\mathrm{top}}(X)$$

is an isomorphism. It follows that j is an isomorphism if and only if $A(X)$ is generated by its elements of degree 1, or, when $k = \mathbb{C}$, if and only if the integral cohomology of $X(\mathbb{C})$ is generated by its elements of degree 2, i.e. if the classifying space $B_{G(\mathbb{C})}$ is without torsion. It is known [15, p. 217, Theorem B, (1) $\Leftrightarrow$ (2), and 2.5] that this holds if and only if G is without torsion, equivalently if G is isomorphic to a product of groups $\mathrm{SL}(n)$ and $\mathrm{Sp}(m)$. Thus it suffices to take for G a group $\mathrm{SO}(n)$ with $n \geq 7$, or a simple exceptional group, to obtain an example in which j is not an isomorphism.

4.6. In this example there can be no doubt that the good ring is the torsion-free ring

$$A(X) \simeq \mathrm{Gr}^{\bullet}_{\mathrm{top}}(X),$$

whose basis consists of the cells in the Bruhat decomposition and which, when $k = \mathbb{C}$, is also isomorphic to the integral cohomology of $X(\mathbb{C})$. By contrast, $\mathrm{Gr}^{\bullet}(X)$ appears as a ring having torsion of a rather pathological nature. This example therefore confirms the view that the invariant $\mathrm{Gr}^{\bullet}(X)$ is hardly interesting

except modulo torsion, i.e. after tensoring with $\mathbb{Q}$. At the same time, this example shows that $\mathrm{Gr}^{\bullet}(X)$ is covariant with respect to proper morphisms only after tensoring with $\mathbb{Q}$, even if one restricts to closed immersions of smooth quasi-projective schemes. Indeed, if X is such that j is not an isomorphism, there is an integral closed subscheme Y of X such that

$$\mathrm{cl}^\bullet(Y) \in K^\bullet(X)$$

does not lie in $\mathrm{Filt}^i(X)$, where $i = \mathrm{codim}(Y, X)$. We may choose Y of minimal dimension, so that

$$\mathrm{Filt}^{i+1}(X) = \mathrm{Filt}^{i+1}_{\mathrm{top}}(X).$$

If Z is the singular locus of Y, and $X' = X \setminus Z$, $Y' = Y \setminus Z$, it follows that

$$\mathrm{cl}^\bullet(Y') \in K^\bullet(X')$$

does not lie in $\mathrm{Filt}^i(X')$. Here one uses the fact that $K^\bullet(X) \to K^\bullet(X')$ is surjective and compatible with the augmentations, and that its kernel is

$$\mathrm{Im}(K(Z) \longrightarrow K(X)) \subset \mathrm{Filt}^{i+1}(X) = \mathrm{Filt}^{i+1}_{\mathrm{top}}(X).$$

Now consider the inclusion

$$f : Y' \longrightarrow X',$$

which is a closed immersion of codimension i between smooth quasi-projective schemes over k, *such that*

$$f_* : K^\bullet(Y') \longrightarrow K^\bullet(X')$$

is not compatible with the λ-filtrations, even up to a shift by i, since $f_*(1) \notin \mathrm{Filt}^i(X')$.

4.7. Retaining the notation of 4.5, note that Exposé IX shows that Hodgkin's theorem quoted above is equivalent to the isomorphism

$$K^\bullet(G) \simeq \mathbb{Z}, \qquad \text{hence} \qquad \mathrm{Gr}^\bullet_{\mathrm{top}}(G) \simeq \mathbb{Z}.$$

The homomorphism $K^\bullet(G/B) \to K^\bullet(G)$ is a priori surjective, and its kernel is the ideal generated by the elements $\mathrm{cl}^\bullet(L) - 1$, where L ranges over the invertible sheaves on $X = G/B$ associated with representations $B \to \mathbb{G}_m$ (which in fact yield all invertible sheaves on X [5, Exposé 15]). An analogous argument shows that

$$A(G/B) \longrightarrow A(G)$$

is also surjective, and that its kernel is the ideal generated by the elements of $A^1(G/B)$ defined by representations $B \to \mathbb{G}_m$, hence by all of $A^1(G/B)$.[3] Thus $A(G)$ is reduced to $A^0(X) = \mathbb{Z}$ if and

Source notes. The scan supplies and controls the references, including [13], no. 6, [1], [4], [RRR], [9], SGA 5 VII, [2], [11], [12], and [15]. It consistently prints a star in $H^{2*}(X, \mathbb{Z}_\ell(*))$, rather than the round graded-ring bullet; that distinction is retained here. In 4.1 the scan associates $\mathrm{Gr}^\bullet(X)$ with the λ-filtration, whereas the locator says γ-filtration; the English follows the scan under `SGA6-XIV-IDX686-LAMBDA-VS-GAMMA-SRCDEF-001`. The scan also prints the full operator $\mathrm{Filt}_{\mathrm{top}}$ in the topological-filtration quotient; the English follows it under `SGA6-XIV-IDX686-FILTTOP-VS-FILTOP-SRCDEF-001`. At the start of 4.2, the scan prints $K^\bullet(X) \to \mathrm{Ch}(A(X))$, whereas the inherited French locator has $K_\bullet(X)$; the English follows the scan under `SGA6-XIV-IDX687-KBULLET-VS-KSUBBULLET-SRCDEF-001`. Earlier in 4.1, the scan likewise prints $K^\bullet(X)$ in both topological-filtration terms, where the locator has $K_\bullet(X)$; the English follows the scan under `SGA6-XIV-IDX686-FILTOP-KBULLET-VS-KSUBBULLET-SRCDEF-001`. Immediately after (4.5), the scan prints that $\mathrm{cl}(Y)$ is the class of Y in $A^i(Y)$. Since Y is a codimension-i cycle on X, and the very

[3] Cf. IX 3.4, 2.2.

next sentence invokes $A^i(X)$, the English uses $A^i(X)$ and records the correction as `SGA6-XIV-IDX688-AIY-VS-AIX-SRCDEF-001`.

Across idx684-idx685 the scan opens but does not close the outer parenthesis around the purity clauses and later repeats k after *characteristic zero*. The English regularizes both under `SGA6-CODEX-663-684-07` and `SGA6-XIV-IDX685-DUPLICATE-K-SRCDEF-001`.

In 3.3 the scan reads "naive K'", rather than the locator's K^0; the English follows the scan under `SGA6-XIV-IDX685-KPRIME-VS-KZERO-SRCDEF-001`.

After (4.6) the scan says that the formula is valid in $\mathrm{Gr}^{\bullet}_{\mathrm{top}}(X)$, although both $f^*f_*(y)$ and y lie on Y; the English uses $\mathrm{Gr}^{\bullet}_{\mathrm{top}}(Y)$ under `SGA6-XIV-IDX689-GTOPX-VS-GTOPY-SRCDEF-001`. On idx690 the scan twice refers to the ℓ-adic analogue of (4.3), where context requires (4.8); the English uses (4.8) under `SGA6-XIV-IDX690-XREF43-VS-XREF48-SRCDEF-001`. On idx692 the scan prints the tautology $\mathrm{Filt}^{i+1}_{\mathrm{top}}(X) = \mathrm{Filt}^{i+1}_{\mathrm{top}}(X)$; the preceding sentence supplies the intended left side $\mathrm{Filt}^{i+1}(X)$, used here under `SGA6-XIV-IDX692-FILTTOP-TAUTOLOGY-SRCDEF-001`.

The scan-exact $\mathrm{cl}(Y)$ and $\mathrm{cl}^{\bullet}(Y)$ notation, the checked $\underline{F}$ in (4.8), and all page-seam citations replace losses in the inherited English candidate. The scan's $\deg(Y \cdot \xi^i) > 0$, its use of $\mathrm{SO}(n)$ after a simply-connected hypothesis, and its $A^0(X) = \mathbb{Z}$ at the idx692 seam are retained without silent emendation. The admitted French authority and Claude's mutable workpass remain unchanged.